Modelling of Advective-Diffusive-Reactive Flows in Geological Carbon Sequestration

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Declaration

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Abstract

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1 Introduction

The solubility trapping of geologically sequestered CO_2 represents one of the most secure, long-term solutions to underground geological carbon storage - second only to the much slower process of mineral trapping - because unlike CO_2 in liquid phase, brine containing dissolved CO_2 becomes denser and thus does not present any risk of rising by buoyancy forces to the top of the geological formation and escaping through structural faults in the cap rock. Convective dynamics driven by such density contrasts have been well-studied in the literature on porous media flows [**?**, **?**]. However, the solubility of CO_2 in water is relatively low and the timescale for the onset of convection processes is estimated to be large [**?**]. One tier down from solubility trapping in the hierarchy of storage security [**?**] (see Figure 1.1) is capillary trapping, whereby small amounts of CO_2 are left behind in the wake of a migrating CO_2 current. The interplay between capillary and solubility trapping provides the motivation for the research forming this thesis. As small bubbles of CO_2 form and are left trapped in pore spaces, nearby brine is expected to partially dissolve these bubbles and become saturated. If the saturated brine is then replaced by unsaturated brine, due to mechanisms such as convection, exchange flow or background hydrogeologic flow, the capillary-trapped CO_2 would further diminish by dissolution, and potentially dissolve entirely.

1.1 One

1.2 Two

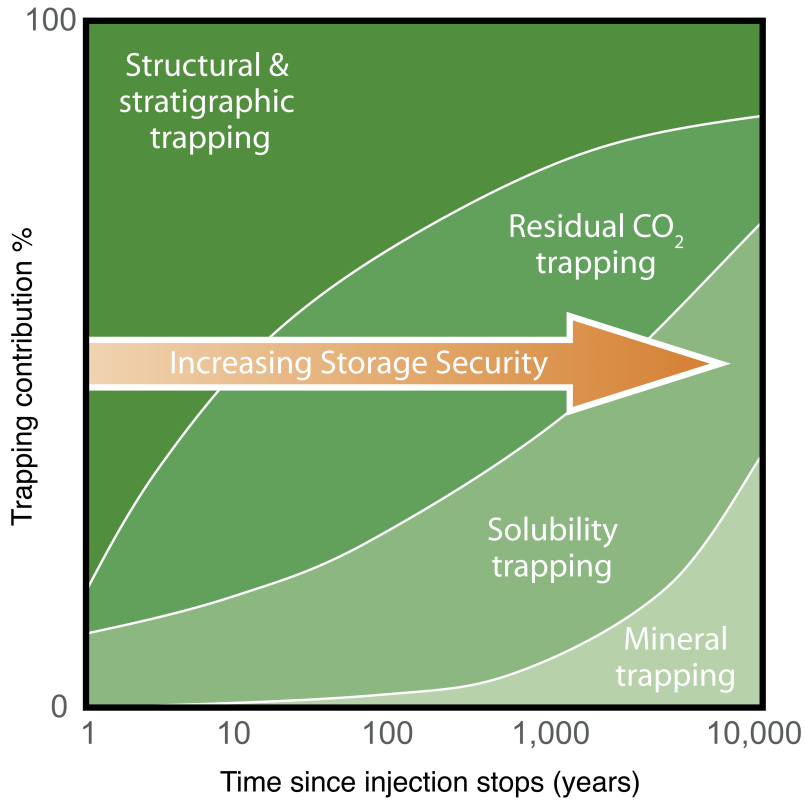

Figure 1.1: Mechanisms of geological CO_2 storage, reproduced from (**metz_ipcc_2005?**).

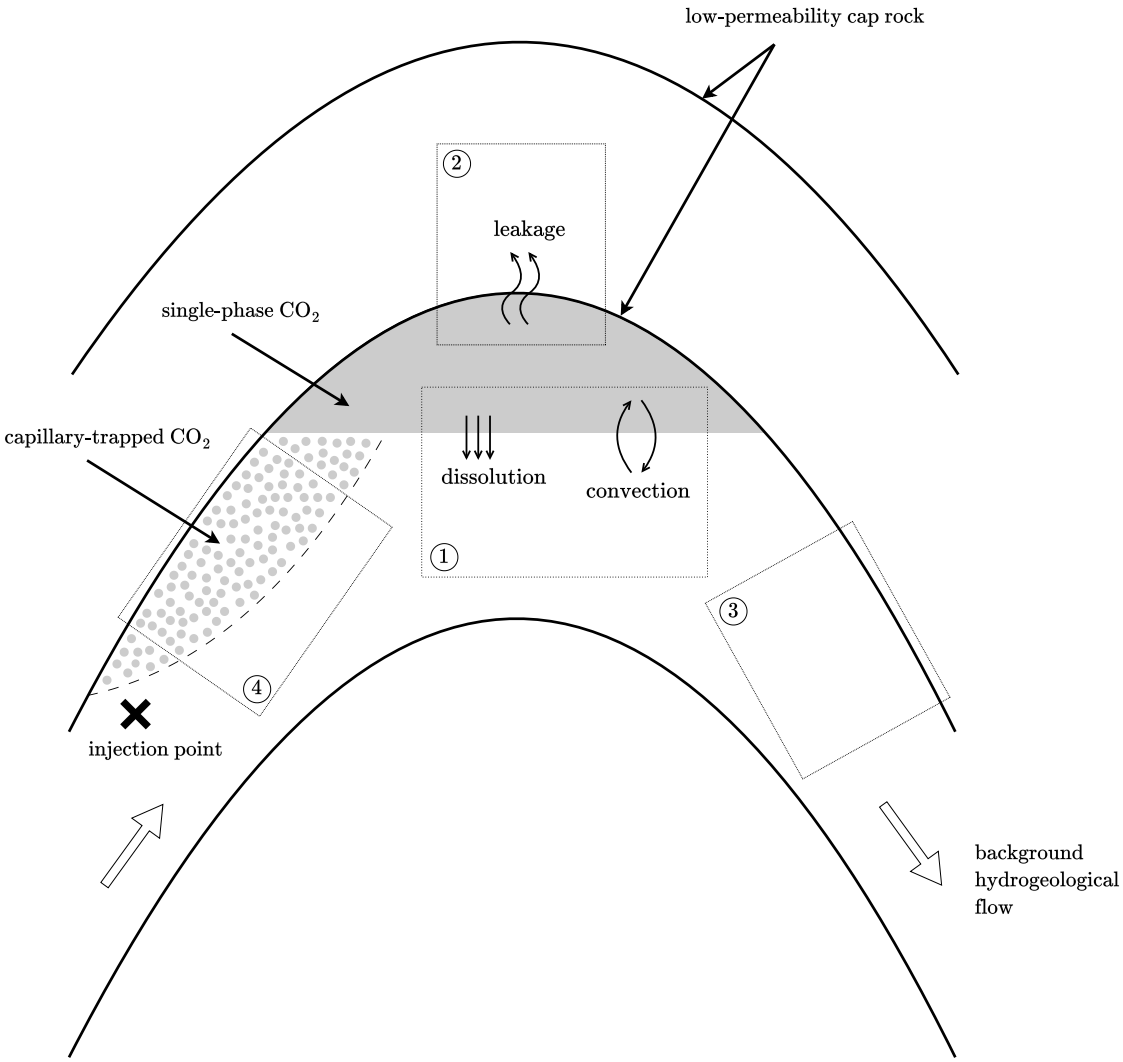

Figure 1.2: Schematic of geologically-sequestered CO_2 in an anticline formation. Single-phase CO_2 is injected at a point downslope of the apex, but rises towards the apex because it is more buoyant than the surrounding brine. Capillary-tragged CO_2 is left behind in its wake. Further rising of the buoyant CO_2 is halted by the impermeable cap rock layer, though leakage may still occur through faults, or over a much longer time scale if the cap rock has low but non-zero permeability. Some anticline formations may also have a background hydrogeological flow from on side to the other. Boxes 1, 2 and 3 isolate the different regions and flows within the anticline formations that research into the fluid dynamics of carbon sequestration has investigated so far with simplified mathematical and numerical models. A simplified model for box 4 is proposed in the following section.

2 Governing equations for the dissolution of capillary-trapped CO_2

2.1 Porosity & saturation

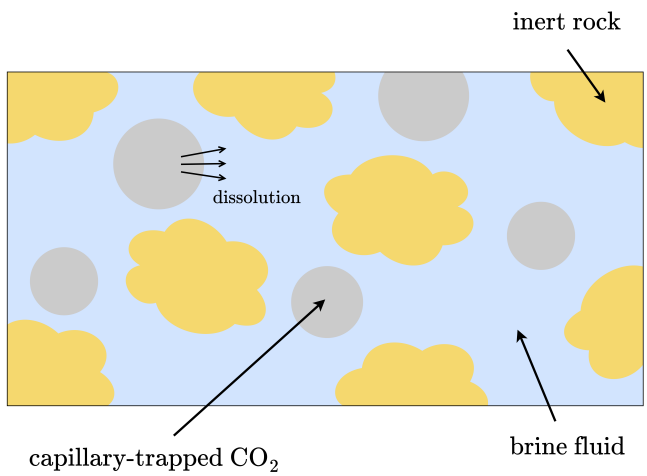

Figure 2.1: Pore-scale schematic illustrating the partition of space into brine fluid, capillary-trapped CO_2 and rock. 'Bubbles' or ganglia of CO_2 are held in place by surface tension forces, but dissolve into the surrounding brine. The rock is assumed inert, reacting with neither CO_2 nor brine.

We consider a porous medium consisting of rock, brine fluid and capillary-trapped CO_2 as illustrated in Figure 2.1 , such that the partition of space can be expressed as

$$\phi + \phi_{CO_2} + \phi_{\text{rock}} = 1 \tag{2.1}$$

where the porosity ϕ is the volume fraction of brine fluid and the remaining volume fractions ϕ_{CO_2} and ϕ_{rock} are contributed by the capillary-trapped CO_2 and rock, respectively. As modelling assumptions we assume that the rock is inert and the 'bubbles' of capillary-trapped CO_2 held in place by surface tension forces are immobile, so that only the brine fluid is

mobile. However the amount capillary-trapped CO_2 is able to gradually diminish over time by dissolution into the neighbouring brine. In general, brine and single-phase CO_2 are a pair of immiscible liquids that can be modelled as a multiphase flow, but in this thesis we only investigate the considerably simpler problem of the single-phase flow of brine fluid containing dissolved CO_2. Given that we are primarily interested in the spatial extent and temporal evolution of the capillary-trapped CO_2, we can write the effective porosity

$$\phi(\mathbf{x}, t) = \varphi(\mathbf{x})(1 - s(\mathbf{x}, t)) \tag{2.2}$$

in terms of the saturation $s(\mathbf{x}, t)$ of capillary-trapped CO_2 and the porosity distribution $\varphi(\mathbf{x})$ of the inert rock formation, which is in general heterogeneous. Since the rock is by assumption inert, $\frac{\partial \varphi}{\partial t} = 0$ and any evolution in the effective porosity is solely due to evolution in the saturation. $s = 0 \Leftrightarrow \phi = \varphi$ corresponds to the pore space containing only brine fluid CO_2, whereas at the other extreme $s = 1 \Leftrightarrow \phi = 0$ corresponds to pore space containing only capillary-trapped CO_2. In term of s and φ, the the other volume fractions are $\phi_{CO_2}(\mathbf{x}, t) = \varphi(\mathbf{x})s(\mathbf{x}, t)$ and $\phi_{\text{rock}}(\mathbf{x}) = 1 - \varphi(\mathbf{x})$.

2.2 Governing equations

In a porous medium filling a domain Ω with boundary $\partial\Omega$, flow coupled to thermosolutal transport is modelled by the general governing equations for $(\mathbf{x}, t) \in \Omega \times [0, \infty)$

$$\phi \frac{\partial \theta}{\partial t} + \nabla \cdot (\theta \mathbf{u} - \mathsf{G} \cdot \nabla c) = 0 \tag{2.3}$$

$$\frac{\partial (\phi c)}{\partial t} + \nabla \cdot (c\mathbf{u} - \mathsf{D} \cdot \nabla c) = R \tag{2.4}$$

$$\frac{\partial (\phi \rho)}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = R \tag{2.5}$$

$$\mathbf{u} = -\frac{\mathsf{K}}{\mu} \cdot (\nabla p - \rho \mathbf{g}) \tag{2.6}$$

$$\varrho \frac{\partial \phi}{\partial t} = R \tag{2.7}$$

where the transport equation (2.3) for the fluid temperature θ with a thermal dispersion G is derived from the conservation of energy (assuming no heat transfer between the solid and liquid phases in the porous medium and no heat sources or sinks); (2.4) describes the transport of solute dissolved at concentration c (solute mass per unit fluid volume) in fluid flowing through a porous medium of porosity ϕ, incorporating advective transport due to the Darcy velocity $\mathbf{u}$, dispersion with coefficient D and rate of addition or removal R due to chemical reactions;

(2.5) expresses mass conservation in the fluid of density ρ; (2.6) is Darcy's law expressing the Darcy velocity $\mathbf{u}$ in terms of the permeability K, fluid viscosity μ, pressure gradient ∇p and buoyancy force $\rho\mathbf{g}$; (2.7) also expresses mass conservation and models the evolution of porosity ϕ due to the addition or removal of solute mass from an immobile phase of density ϱ. Note that the Darcy velocity differs from the interstitial fluid velocity $\phi\mathbf{u}$ by a factor of porosity. This system of equations is closed by a set empirical relations for the concentration-dependent fluid density $\rho = \rho(c)$ and viscosity $\mu = \mu(c)$ (i.e. equations of state), porosity-dependent permeability $\mathsf{K} = \mathsf{K}(\phi)$, dispersions $\mathsf{G} = \mathsf{G}(\phi, \mathbf{u})$ and $\mathsf{D} = \mathsf{D}(\phi, \mathbf{u})$ dependent on both porosity and Darcy velocity, and a reaction rate $R = R(\phi, c, \theta)$ dependent on porosity, concentration and temperature. G, D and K are in general tensor-valued to account for anisotropy in the porous medium. The gravity vector expressed in terms of its unit vector is $\mathbf{g} = g\mathbf{e}_g$.

A specific scenario solving the governing equations (2.3)-(2.7) is supplemented by a set of boundary conditions on θ, c, and ($\mathbf{u}$, p) and initial conditions on θ, c and s. In general the boundary conditions on the transport equations (2.3) and (2.4) may be a combination of Dirichlet (D) and Neumann (N) boundary conditions prescribing a known value or flux respectively

$$\theta = \theta_{\mathrm{D}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{D},\theta} \tag{2.8}$$

$$\mathbf{n} \cdot (\mathsf{G} \cdot \nabla\theta) = \theta_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N},\theta} = \partial\Omega / \partial\Omega_{\mathrm{D},\theta} \tag{2.9}$$

$$c = c_{\mathrm{D}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{D},c} \tag{2.10}$$

$$\mathbf{n} \cdot (\mathsf{D} \cdot \nabla c) = c_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N},c} = \partial\Omega / \partial\Omega_{\mathrm{D},c} \tag{2.11}$$

where $\mathbf{n}$ is the outward unit normal to the boundary. The boundary conditions on the fluid flow governed by the Darcy system (2.5) and (2.6) may be either a prescribed pressure, or a prescribed normal velocity. For reasons that will become evident later in our discussion of the finite element method, we shall refer to these as essential (E) and natural (N) boundary conditions respectively.

$$\mathbf{n} \cdot \mathbf{u} = u_{\mathrm{E}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{E}} \tag{2.12}$$

$$p = p_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N}} = \partial\Omega / \partial\Omega_{\mathrm{E}} \tag{2.13}$$

3 Numerical software

References